

Special Relativity: The Michelson–Morley Experiment and the Consequences of the Postulates

PHYS 2203 — Introductory Physics III

September 21, 2026

Overview

This lecture marks the transition from optics to relativity. After introducing the ideas of an *inertial frame* and an *event*, we confront a puzzle raised by Maxwell’s equations. Our wave equations give light a definite speed c , but relative to no particular frame. The proposed answer — a medium called the ether — is tested by the Michelson–Morley experiment, whose null result shows that light needs no medium and has no special frame. This forces Einstein’s two postulates of special relativity, from which we draw the first two consequences for space and time — the relativity of *simultaneity* and *time dilation* — and we close by posing the *twin paradox*.

1 Reference Frames and Events

All of relativity rests on two basic notions — the *events* we observe and the *inertial reference frames* from which we observe them. We define both carefully before going further.

Event. An *event* is something that happens at a single point in space at a single instant in time. It is specified by four numbers: three spatial coordinates and one time, (x, y, z, t) . A firecracker exploding or two particles colliding at one place and one moment are events.

Inertial reference frame. An *inertial reference frame* is a coordinate system in which Newton’s first law holds: an object with no net force on it moves in a straight line at constant velocity. Any frame moving at constant velocity relative to an inertial frame is itself inertial, while a frame that accelerates or rotates is not.

Concretely, a frame is a set of coordinate axes with a clock at every point, carried by some observer. Two observers in uniform relative motion — an unprimed frame O and a primed frame O' moving at constant velocity $\mathbf{v}$ — each assign coordinates (x, y, z, t) to the same event, as shown below.

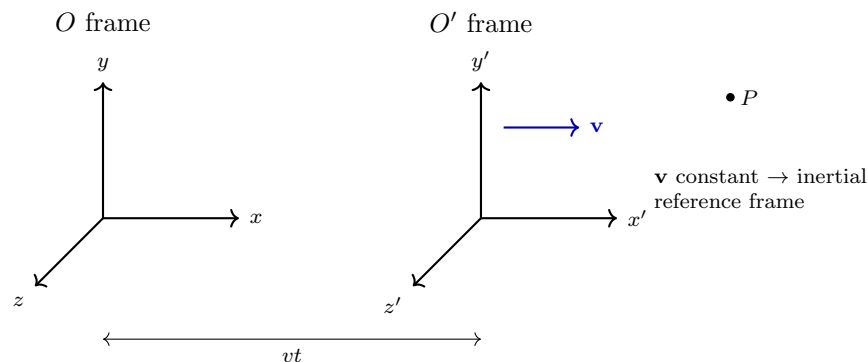


Figure 1: An event at point P described by two inertial observers. Frame O' moves with constant velocity $\mathbf{v}$ along x , so after a time t its origin has shifted by vt relative to O .

2 The Speed of Light: Which Frame?

Maxwell's equations predict that light travels through empty space at a single, definite speed,

$$c \approx 3.00 \times 10^8 \text{ m/s},$$

but they single out no particular reference frame. Speed c *relative to what?*

Every other wave we know travels at a fixed speed relative to a **medium**, not relative to its source. Sound is a pressure disturbance in air, and it moves at about 340 m/s relative to the air, whatever the source is doing. Ripples move at a fixed speed relative to the water. By analogy, nineteenth-century physicists supposed that light also needed a medium — an invisible substance filling all of space, the **luminiferous ether**. Light would travel at c relative to the ether, and the ether would supply the universal rest frame that Maxwell's equations seemed to demand.

If the ether exists, the Earth's motion through it should produce an **ether wind**, so the measured speed of light should depend on direction — fastest with the wind, slowest against it. Detecting that difference is exactly what the Michelson–Morley experiment set out to do.

3 The Michelson–Morley Experiment

The Michelson–Morley apparatus is the **Michelson interferometer**, which splits a beam of light into two perpendicular arms, reflects each off a mirror, and recombines them. The recombined beams interfere, and the pattern is exquisitely sensitive to any difference in the two round-trip travel times.

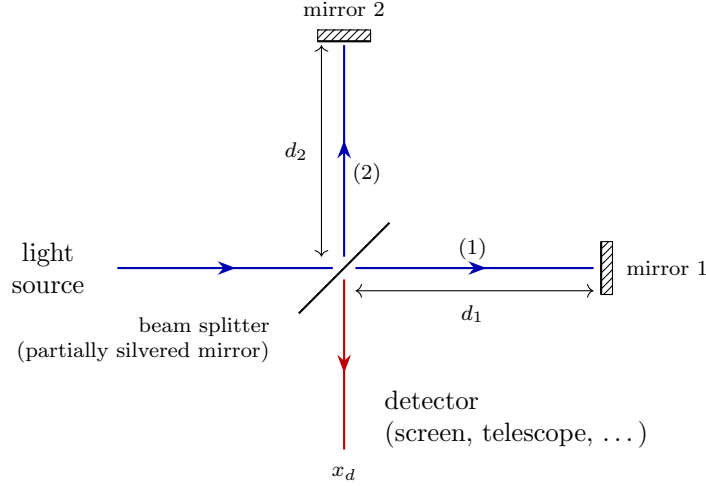


Figure 2: The Michelson interferometer. The beam splitter sends beam (1) to mirror 1 (distance d_1) and beam (2) to mirror 2 (distance d_2). The reflected beams recombine and travel to the detector.

The phase difference between rays (1) and (2) at the detector is

$$\delta = \frac{2\pi}{\lambda} (2d_2 - 2d_1). \quad (1)$$

Each beam traverses its arm twice, hence the factors of $2d_1$ and $2d_2$. Moving one mirror by only $\lambda/2$ changes a round-trip path by λ and sweeps one full bright–dark fringe past the detector, so the instrument detects path-length differences far smaller than a wavelength of light.

The ether-wind hypothesis. In the language of Figure 1, our apparatus is the frame O and the ether is a frame O' streaming past it at speed v . If the apparatus moves to the right at speed v through the stationary ether, an observer riding with it feels an **ether wind** blowing to the left at speed v . Take the two arms to have equal length L and ask how long the light takes to make the round trip along each.

Path ①: along the direction of motion

For the beam that travels parallel to the motion, the trip out (downstream, with the wind) is at effective speed $c + v$ and the return (upstream, against the wind) is at $c - v$. The round-trip time over a one-way length L is

$$t_1 = \frac{L}{c + v} + \frac{L}{c - v} = \frac{2Lc}{c^2 - v^2} = \frac{2L}{c} \frac{1}{1 - v^2/c^2}. \quad (2)$$

Path ②: perpendicular (cross-stream)

For the beam that travels perpendicular to the motion, the effective speed is *not* c , and the reason is worth drawing.

Work in the ether frame, where light really does travel at c in a straight line. During the time t it takes the light to reach mirror 2, the apparatus drifts sideways by vt . If the beam were aimed

straight across it would miss the mirror entirely, so it must be aimed slightly *upstream*. Its actual path through the ether is the slanted line of length ct in Figure 3.

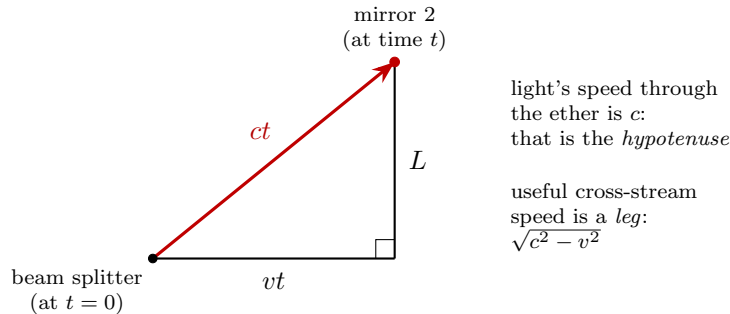


Figure 3: The cross-stream leg in the ether frame. The light must be aimed upstream to land on the drifting mirror, so its path of length ct is the hypotenuse of a right triangle whose legs are the sideways drift vt and the arm L .

We then have,

$$(ct)^2 = L^2 + (vt)^2 \quad \implies \quad t^2 (c^2 - v^2) = L^2 \quad \implies \quad t = \frac{L}{\sqrt{c^2 - v^2}}, \quad (3)$$

so the effective speed *across* the stream is $\sqrt{c^2 - v^2}$. The round-trip time is then

$$t_2 = \frac{L}{\sqrt{c^2 - v^2}} + \frac{L}{\sqrt{c^2 - v^2}} = \frac{2L}{\sqrt{c^2 - v^2}} = \frac{2L}{c} \frac{1}{\sqrt{1 - v^2/c^2}}. \quad (4)$$

Since these times are different ($t_1 \neq t_2$), an ether wind would put the two beams out of step. Rotating the whole apparatus by 90° swaps the two arms and should shift the interference fringes by a predictable, easily measurable amount.

The null result. No shift was ever seen. Michelson repeated and refined the experiment for decades, always with the same outcome: the two round-trip times are equal, whatever the apparatus's orientation or motion. There is no ether wind, and no special “ether frame” for light. Light, unlike sound or water waves, requires *no medium at all* — it propagates through empty space, and its speed comes out the same in every inertial frame.

4 Einstein's Theory of Special Relativity

Einstein reached the same conclusion — there is no ether — much earlier, and from a completely different direction. Rather than patch up classical physics, he raised two simple statements to the status of **postulates** and followed them wherever they led.

The two postulates

1. **The principle of relativity:** The laws of physics are the same in all inertial reference frames.
2. **The principle of the constancy of the speed of light:** The speed of light in free space has the same value c in all inertial reference frames.

Postulate 2 leads to revolutionary changes in the concepts of space and time. In this lecture we draw the first two consequences — the relativity of simultaneity and time dilation — and then meet a puzzle they raise, the twin paradox. A third consequence, length contraction, follows next lecture.

5 Consequence I: Relative Simultaneity

Consider two observers, Anna and Bob. Anna rides a flatcar moving east at speed v , holding a flashbulb at each end of the car. She fires both at the same instant *in her frame*. Bob stands on the ground and watches her pass. The two flashes are two *events* that happen at different locations.

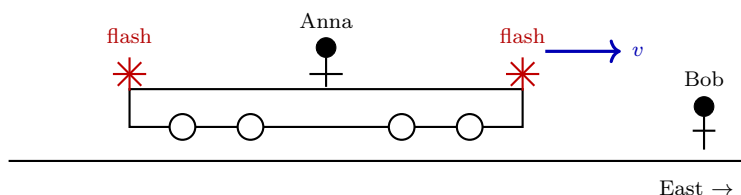


Figure 4: Anna, on a flatcar moving east at speed v , fires a flashbulb at each end at the same instant in her frame. Bob watches from the ground.

In Anna's frame. The two flashbulbs are equal distances from her, and by the second postulate both beams travel toward her at the same speed c . The beams reach her at the same instant, so in Anna's frame the two flashes are *simultaneous*.

In Bob's frame. While the light is travelling, Anna moves east. The beam from the *west* (rear) bulb must chase her and cover a longer distance, while the beam from the *east* (front) bulb has a shorter distance to cover. Both observers must agree that the two beams reach Anna at one and the same event. But if in Bob's frame the west beam travels farther yet still arrives at the same moment, then in Bob's frame the west bulb must have fired *first*. For Bob the two flashes are therefore *not* simultaneous.

Relativity of simultaneity. Two events at different locations that are simultaneous in one inertial frame are *not* simultaneous in a frame moving relative to the first.

This is not an optical illusion caused by the travel time of light to Bob's eyes. Bob could post an assistant right next to each flashbulb, each recording the local time at which the nearby bulb fires. They would still record two different times. The effect is real, though far too small to notice at everyday speeds.

6 Consequence II: Time Dilation

Consider a **light clock**: a light source and detector at the bottom, a mirror a distance L_0 above, and a pulse of light that bounces up and back. We first analyze it at rest in the O' frame.

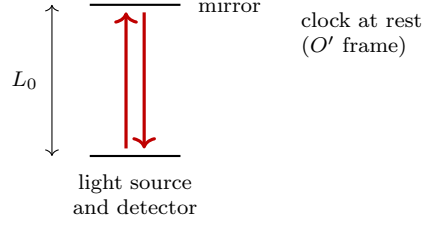


Figure 5: A light clock at rest in the O' frame. Light leaves the source, reflects off the mirror a height L_0 above, and returns to the detector.

Proper time. One tick of the light clock takes

$$\Delta t_0 = \frac{2L_0}{c}. \quad (5)$$

This is the **proper time**, measured when the clock is at rest. Δt_0 is the time interval between two events that occurred at the *same place*:

- Event 1: light emitted.
- Event 2: light detected.

The moving clock (the O frame)

In the O frame, the clock is moving to the right with speed v . During one tick, the light must travel along the two slanted sides of a triangle, because the mirror and detector move horizontally while the light is in flight.

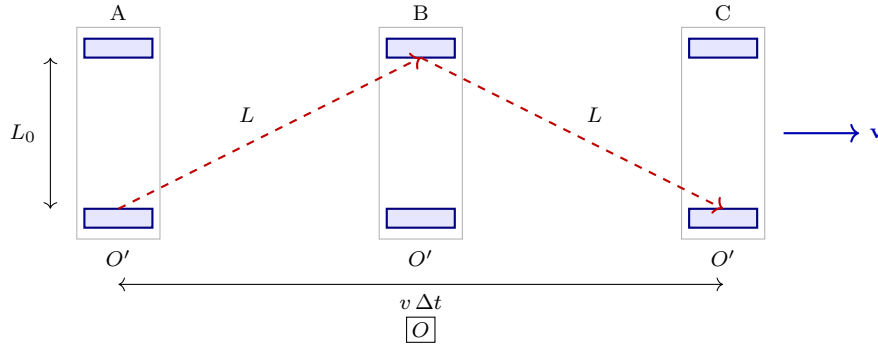


Figure 6: The same light clock seen from the O frame, in which it moves to the right at speed v . The light travels along the slanted hypotenuses (each of length L) while the clock advances a horizontal distance $v \Delta t$ during one tick.

Geometry of one tick. The light's total path length equals its speed times the elapsed time. Using distance travelled = speed $\times$ time and the Pythagorean theorem for each slanted leg,

$$\Delta t = \frac{2L}{c} = \frac{2}{c} \sqrt{\left(\frac{1}{2}v \Delta t\right)^2 + L_0^2}. \quad (6)$$

Squaring,

$$\Delta t^2 = 4 \left(\frac{1}{2} \frac{v}{c} \right)^2 \Delta t^2 + \frac{4L_0^2}{c^2}. \quad (7)$$

Recognizing $4L_0^2/c^2 = (2L_0/c)^2 = \Delta t_0^2$ and gathering the Δt^2 terms (the second postulate, c the same in both frames, is used here):

$$\Delta t^2 \left(1 - \frac{v^2}{c^2} \right) = \Delta t_0^2 \quad \Rightarrow \quad \boxed{\Delta t = \frac{\Delta t_0}{\sqrt{1 - v^2/c^2}}} \quad (\Delta t > \Delta t_0). \quad (8)$$

Δt is the **dilated time**, which is measured when the two events occur in *different places*. Because the denominator is less than 1, a moving clock runs slow: $\Delta t > \Delta t_0$.

The Lorentz factor

This combination appears in every result from here on, so we give it a name.

The Lorentz factor. $\gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}$, where v is the relative speed of the two frames. Since $v < c$ we always have $\gamma \geq 1$, with $\gamma \rightarrow 1$ when $v \ll c$ and $\gamma \rightarrow \infty$ as $v \rightarrow c$.

In terms of γ the time-dilation result is simply

$$\boxed{\Delta t = \gamma \Delta t_0} \quad (9)$$

and from now on we write γ rather than the square root. A few values worth remembering:

v/c	0.10	0.50	0.80	0.90	0.99	0.999
γ	1.005	1.15	1.67	2.29	7.09	22.4

At everyday speeds γ is indistinguishable from 1, which is why none of this was noticed for centuries.

Time dilation. Two events that occur at the *same location* in one inertial frame are separated by a *longer* time interval when measured in a frame moving relative to the first.

Generality and example

This equation is true for the time interval between *any* two events.

Event 1	Event 2
spaceship leaves Earth	spaceship arrives at Mars
clock ticks	clock ticks again
heart beats	heart beats again

Whether one counts clock ticks, heartbeats, or the journey of a spaceship, the proper-time interval Δt_0 measured by the moving traveler is shorter than the dilated interval Δt measured in the frame where the two events occur at different places.

7 A Puzzle: The Twin Paradox

Time dilation leads at once to a famous puzzle. Suppose twins Alice and Bob begin together on Earth. Alice boards a rocket, travels to a distant star at a speed close to c , turns around, and flies home, while Bob stays behind. Because Alice's clock runs slow in Bob's frame, Bob expects Alice to come back *younger* than himself.

Here is the trouble. Motion is relative, so from Alice's point of view it is Bob who rushes away at high speed and later returns. Each twin sees the *other's* clock running slow, so the situation looks perfectly symmetric. How, then, can one of them end up definitely younger? They cannot each be younger than the other when they are reunited.

The twin paradox. One twin travels at high speed and returns to find the stay-at-home twin has aged more. Because each twin sees the other's clock run slow, it is not obvious which twin should end up younger — or whether there should be any difference at all.

The way out is that the two twins are *not* in symmetric situations. Bob stays in a single inertial frame the whole time, while Alice must turn around, so she is accelerated and spends her trip in two different inertial frames. Turning this observation into a definite answer — and showing that Alice really does return younger — requires tools we build over the next few lectures, in particular the Lorentz transformation and the spacetime diagram. We will return to the twin paradox and resolve it completely then.